(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
15 August 2024 (15.08.2024)

WIPO | PCT

(10) International Publication Number
WO 2024/165898 A1

(51) International Patent Classification:
A61M 5/145 (2006.01) *A61D 7/00* (2006.01)

(21) International Application Number:
PCT/IB2023/053871

(22) International Filing Date:
15 April 2023 (15.04.2023)

(25) Filing Language: English

(26) Publication Language: English

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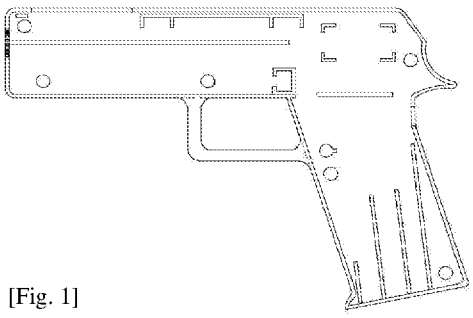
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(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MU, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, CV, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SC, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:
— *with international search report (Art. 21(3))*
— *before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))*
— *upon request of the applicant, before the expiration of the time limit referred to in Article 21(2)(a)*

(54) **Title:** AUTOMATED DIGITAL INJECTION GUN WITH THE ABILITY TO STORE THE INJECTION DATA OF EACH LIVESTOCK.



[Fig. 1]

(57) **Abstract:** An animal injection gun with a comprehensive system of drug and vaccine injection for domestic and wild animals, which, in addition to performing hard injections, stores comprehensive and complete information on the proportion of each animal in the system so that if needed, this information can be used from the birth of the new baby to the end of the animal's life and prevented the spread of herd diseases. The components of this device include: the electronic circuit controlling the engine, the part which functions for injection, the placement location of the carpules, battery, controlling lights, needle head, push button, battery, storage location of injection data and other parts of the device has been embedded on the gun.

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Description

Title of Invention: Automated Digital Injection Gun with the ability to store the injection data of each livestock.

[0001] An animal injection gun with a comprehensive system of drug and vaccine injection for domestic and wild animals, which, in addition to performing hard injections, stores comprehensive and complete information on the proportion of each animal in the system so that if needed, this information can be used from the birth of the new baby to the end of the animal's life and prevented the spread of herd diseases. The components of this device include: the electronic circuit controlling the engine, the part which functions for injection, the placement location of the carpules, battery, controlling lights, needle head, push button, battery, storage location of injection data and other parts of the device has been embedded on the gun.

Technical Field

[0002] A61D7/00; A61M5/145; G16H20/17

Background Art

[0003] Relating to conducting the tests mentioned above, normal human medicine syringes or automated livestock vaccination syringes are used in farms and horse farms. With due attention to the interior design of human medicine syringes for livestock medicine purposes, incorporating them requires great precision when injecting, in terms of the size of the needle head's infiltration to areas such as the lower eyelid area in Mallein Test and injecting the right amount of the effecting drug to the amount of 0/1 CC (The required amount of the effecting drugs in the Mallein test, Tuberculin test, and the Dehorning drug); Injecting the precise amount of it requires extreme binding of the livestock or high promptitude. Automated livestock vaccination syringes have a low precision due to their mechanicalness when used for injections. On the topic of precise injection tools, the automated infusion pumps of drugs are incorporated in human medicine or veterinary anesthesia, but because of the humongous size of the device, the extreme price, its need for urban electricity to function as well as its functioning method (intravenous injection) makes it incapable of being used in the mentioned fields. Furthermore, no similar in-house or foreign sample designed and created for this sample exists.

[0004] Sensing Complete Injection for Animal Injection Device

[0005] United States Patent Application 20210338402

[0006] A semi-automated veterinary medicine delivery device may sense whether and how much air may have been incorporated into a dose delivered to an animal. The device may also sense an incomplete injection in some cases, and may cause a second dose to

be administered to make up for the incomplete dose. The device may compare a position sensor on a syringe in combination with a current sensor on a motor attached to the syringe to detect any abnormalities. In the case of an improper dose, the device may alert the user and administer a second dose.

[0007] Digital sampling apparatus and methods for sorting particles

[0008] United States Patent 8198092

[0009] A system and method for sorting a mixture of stained particles including a digital signal processor for analyzing and classifying the digital information generated from the particles and providing a sorting signal to a sorting system as a function of the analyzed and classified digital information.

Summary of Invention

[0010] To diagnose diseases such as Glanders in horses and/or Tuberculosis in cows, the Mallein Test or Tuberculin Test is routinely conducted by injecting small doses of antigenic substances in the skin or under the skins of livestock in farms and horse farms. Furthermore, due to a lack of collecting and gathering the injection information on each livestock, the places that care for domicile and wild animals have no database on injections that could relay which drugs and vaccines the livestock received and in what way. The goal of this invention is to design and create a comprehensive system of drug and vaccine injection for domicile and wild livestock that in addition to performing difficult injections, could store complete and comprehensive information based on each livestock in the system so that if needed, information ranging from childbirth to the slaughter or the end of an animal's life could be used to prevent herd diseases.

Technical Problem

[0011] To diagnose diseases such as Glanders in horses and/or Tuberculosis in cows, the Mallein Test or Tuberculin Test is routinely conducted by injecting small doses of antigenic substances in the skin or under the skins of livestock in farms and horse farms. Due to the country's woes with the fatal and extremely dangerous Glanders disease and the commonality of this disease with humans, as well as the livestock and financial loss from the Tuberculosis disease in cows, these tests are conducted on a widespread basis every six to twelve months in all the farms and horse farms. Furthermore, by attacking other livestock and ranchers, the existence of horned animals in cow cattle, herds of goats, and industrial and traditional sheep inflicts damages and causes an increase in the costs of care and treatment of livestock. With due attention to the modern methods of dehorning and the new low-volume injection drugs provided in this regard by inventors, the need for a tool capable of rapidly and precisely injecting an appropriate portion of the preceding drugs and conducting the aforesaid tests in the skin area or under the skin (For the mentioned tests, the lower eyelid area in livestock

and for the dehorning drug, in the horn bud area) was felt.

[0012] Furthermore, due to a lack of collecting and gathering the injection information on each livestock, the places that care for domicile and wild animals have no database on injections that could relay which drugs and vaccines the livestock received and in what way; This problem, along with using the traditional method of manually entering the information based on the animal ID Cards could cause issues such as veterinarians not being informed on the vaccines already injected to the livestock or the course of time they served.

[0013] The goal of this invention is to design and create a comprehensive system of drug and vaccine injection for domicile and wild livestock that in addition to performing difficult injections, could store complete and comprehensive information based on each livestock in the system so that if needed, information ranging from childbirth to the slaughter or the end of an animal's life could be used to prevent herd diseases.

Solution of problem

[0014] In designing and building this invention, an electrical engine functioning with the generated electricity of a 9-volt battery is used and courtesy of a controller board in the gun, it controls the amount of the injected drug in a precise and fully programmable way that by pushing the device buttons, it injects the required amount. To give more control to the injector as well as to prevent unwanted and mistaken injections, two buttons have been included in the gun handle so that when both are triggered, the device performs the act of injection. The gun engine is attached to a threaded rod; When the work gets underway, and this rod rotates, the gun engine drives forward the piece that has at its center a nut of the same size; This piece itself is connected to another hollow rod which eventually, by courtesy of pressuring the back of the drug carpules, injects the necessary amount at the injection site. To prevent the simultaneous movement of the mentioned piece, a groove exists on the plastic body of the gun similar to the groove found on the piece as well, causing it to be driven forward and not rotated around itself. For further caution, an on/off button for the device has been embedded on the gun injection handle; there are red lights to inform the user that the device is turned off and yellow lights to inform and caution the user that the device is in injection mode. The handle and outer structure of this device have been built using plastics, following its design using the AutoCAD software, it was created in a highly elegant and resistant way using a 3D printer. There is an area on this handle meant for changing the device battery and placing the carpules of the previously mentioned drugs, as well as its special needle heads that have been embedded from the common double-sided needle heads in the market. Also, for precision in the area and volume of the infiltrating needle to the skin, a plastic part has been placed on the needle head area which prevents further infiltration than the required extent and increases the resistance

of the needle head against the tissue resistance arising from infiltration.

[0015] The gun contains a digital lock that for every Injection, requires the input of each livestock's ID Card, attached to their ears from birth as a certificate by a veterinarian, then the vaccine or drugs information or the type of test is entered, and finally, the system is ready for injection; This information could differ for each farm based on the ID card.

Advantage effects of invention

[0016] Lower cost of conducting the tests thanks to the capacity of continuously using the device courtesy of the ability to change the needle head and carpules instead of replacing the syringe with each injection

[0017] The preciseness of the injection portion

[0018] The preciseness of the extent of needle head infiltration on the skin

[0019] Injection using 2 CC carpules which, considering the need to inject 0/1 CC, with each charging of the carpules, can be used for 20 injections.

[0020] The rapidity and simplicity of the injection

Brief Description of Drawings

[0021] [Fig.1]: Overview of the livestock injection gun

[0022] [Fig.2]: Back and side view of livestock injection gun

[0023] [Fig.3]: Carpool storage compartment

[0024] [Fig.4]: Automatic and digital injection gun cartridge clip

[0025] [Fig.5]: the trigger

[0026] [Fig.6]: The moving part of the automatic and digital injection gun

[0027] [Fig.7]: Carpool entrance cover

[0028] [Fig.8]: Needle storage and connection

[0029] [Fig.9]: battery

[0030] [Fig.10]: push-button

Description of Embodiments

[0031] [Fig.1]: Overview of the livestock injection gun

[0032] [Fig.2]: Back and side view of livestock injection gun

[0033] [Fig.3]: Carpool storage compartment

[0034] [Fig.4]: Automatic and digital injection gun cartridge clip

[0035] [Fig.5]: the trigger

[0036] [Fig.6]: The moving part of the automatic and digital injection gun

[0037] [Fig.7]: Carpool entrance cover

[0038] [Fig.8]: Needle storage and connection

[0039] [Fig.9]: battery

[0040] [Fig.10]: push-button

Examples

[0041] The Technician, the Veterinarian, or the Rancher will place the drug carpules on the gun, they will put the mentioned needle head and tighten it, and will put the injection guider on the needle head, hence by powering the device on, the gun will be ready for injection; After determining the injection area on the livestock based on the type of the test or the drug in use, by holding both injection buttons in hand, they will inject one-tenth of the drug on the skin or under the skin area.

Industrial Applicability

[0042] Mallein Test in horses, Tuberculin Test in cows, Dehorning Drugs in Cows, Sheep, and Goats, as well as the injection of drugs or other substances capable of being distributed in the specialized carpules of this gun and that require a special amount and high injection precision.

Claims

[Claim 1] An animal injection gun with a comprehensive system of drug and vaccine injection for domestic and wild animals, which, in addition to performing hard injections, stores comprehensive and complete information on the proportion of each animal in the system so that if needed, this information can be used from the birth of the new baby to the end of the animal's life and prevented the spread of herd diseases. The components of this device include: the electronic circuit controlling the engine, the part which functions for injection, the placement location of the carpules, battery, controlling lights, needle head, push button, battery, storage location of injection data and other parts of the device has been embedded on the gun.

[Claim 2] According to claim 1, the conduction methods of the Mallein Test, Tuberculin Test, and Dehorning include the injection of Anti-Genes of the pathogenic agents or dehorning drugs that require immaculate precision and quickness and the infiltrating size of the needle head to the injection side is of importance.

[Claim 3] The method of claim 2 that requires the Anti-Genes or drugs be injected in adequate portions (Usually 0/1 CC) in the injection site which for the Mallein Test and Tuberculin Test is in the lower eyelid area of the livestock (Mallein test for horses and Tuberculin test in cattle) and for dehorning (Cattle, Sheep, Goat) is in the horn bud area.

[Claim 4] To implement claim 3, an electric engine was placed in an embedded location inside the gun which works courtesy of a 9-volt battery and by pressuring a part using a dynamic rotating rod, performs the function of injecting drugs or other substances inside the carpules. To be informed about the power status of the gun as well as the enablement of the injection procedure, Two LED lights, one in Red (When the gun is turned on) and the other in Yellow (When the injection procedure is enabled) were placed separately in the gun frame.

[Claim 5] About claim 4, an electronic circuit was placed on the gun frame to control the injection portion, the function, and the movement speed of the electric engine. By using an on/off button, this circuit controls the transferred battery to the gun as well as its activeness or inactivity.

[Claim 6] A button, referred to in claim 5, has been embedded in the gun frame and the electronic circuit that following the injection of the substances within the carpool, brings the injector back to its starting place so the

user could change the carpules.

[Fig. 1]

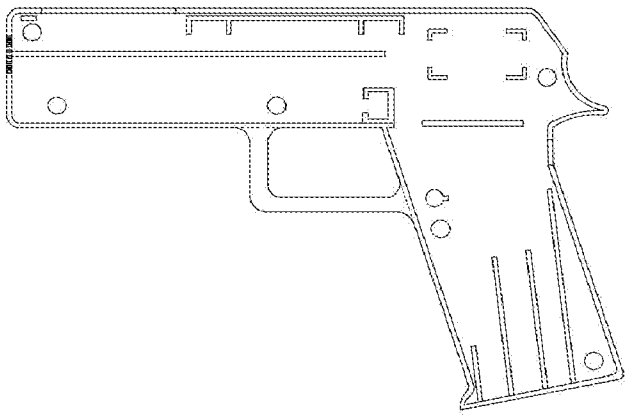


Fig 1

[Fig. 2]

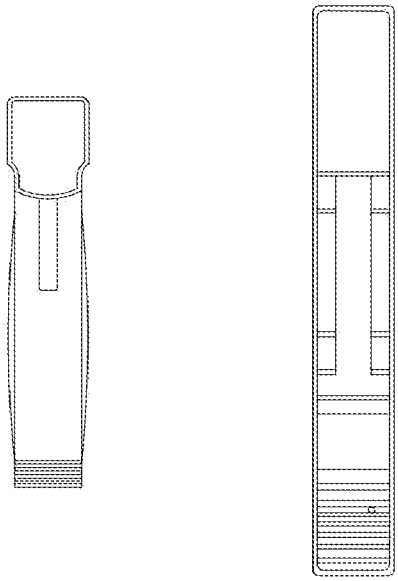


Fig 2

[Fig. 3]

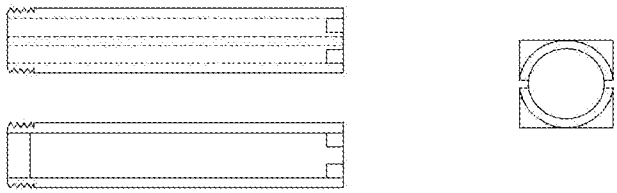


Fig 3

[Fig. 4]

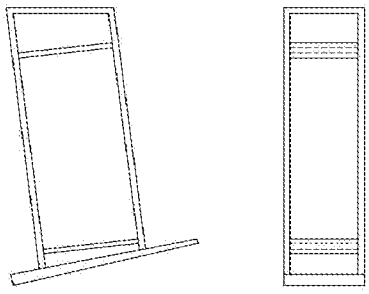


Fig 4

[Fig. 5]

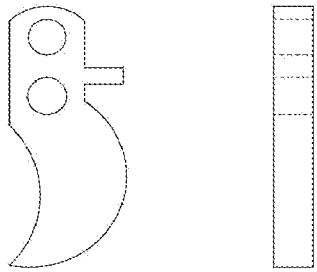


Fig 5

[Fig. 6]



Fig 6

[Fig. 7]

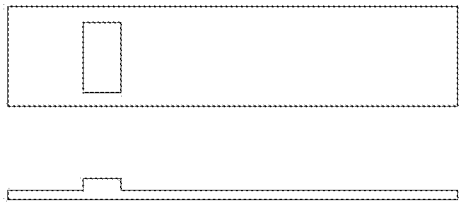


Fig 7

[Fig. 8]

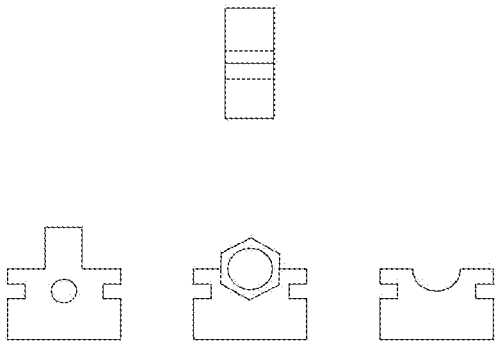


Fig 8

[Fig. 9]

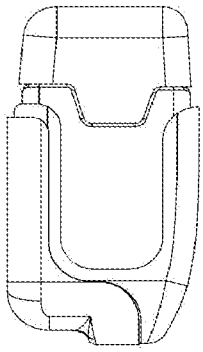


Fig 9

[Fig. 10]

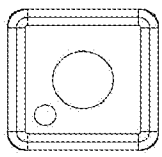


Fig 10

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2023/053871

A. CLASSIFICATION OF SUBJECT MATTER
A61M5/145,A61D7/00 Version=2023.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61M,A61D

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

DATABASES: PatSeer, IPO Internal Database
KEYWORDS: INJECTION GUN,STORE, INJECTION DATA

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US2018344442A1-(DIGIRODEO SA)(06-Dec-2018)(06-12-2018) Claims 1,13,15 and 18, Figure 1, Paragraphs[0001],[0002],[0011],[0047],[0056] and [0069]	1-6
A	CN204468342U-(HUANG CHAO)(15-Jul-2015)(15-07-2015) Whole document	1-6

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
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"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
30-06-2023	30-06-2023
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